EXTERIOR PRODUCT AS ITERATED PULLBACK AND AS INTERSECTION PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0AZM: sources/stacks-chow/tags.txt, source lines 13830–13830. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B02: sources/stacks-chow/tags.txt, source lines 13845–13845. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X and Y be nonsingular varieties. Let $\alpha \in Z_r(X)$ and $\beta \in Z_s(Y)$. Then*

1. *$pr_Y^*(\beta) = [X] \times \beta$ and $pr_X^*(\alpha) = \alpha \times [Y]$,*
2. *$\alpha \times [Y]$ and $[X] \times \beta$ intersect properly on $X \times Y$, and*
3. *we have $\alpha \times \beta = (\alpha \times [Y]) \cdot ([X] \times \beta) = pr_Y^*(\alpha) \cdot pr_X^*(\beta)$ in $Z_{r+s}(X \times Y)$.*

Proof. By linearity we may assume $\alpha = [V]$ and $\beta = [W]$. Then (1) says that $\mathrm{pr}_Y^{-1}(W) = X \times W$ and $\mathrm{pr}_X^{-1}(V) = V \times Y$. This is clear. Part (2) holds because $X \times W \cap V \times Y = V \times W$ and $\dim(V \times W) = r + s$ by Lemma 0AZM.

Proof of (3). Let ξ be the generic point of $V \times W$. Since the projections $X \times W \to W$ is smooth as a base change of $X \to \mathrm{Spec}(\mathbf{C})$, we see that $X \times W$ is nonsingular at every point lying over the generic point of W, in particular at ξ. Similarly for $V \times Y$. Hence $\mathcal{O}_{X \times W, \xi}$ and $\mathcal{O}_{V \times Y, \xi}$ are Cohen-Macaulay local rings and Lemma 0B02 applies. Since $V \times Y \cap X \times W = V \times W$ scheme theoretically the proof is complete. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.